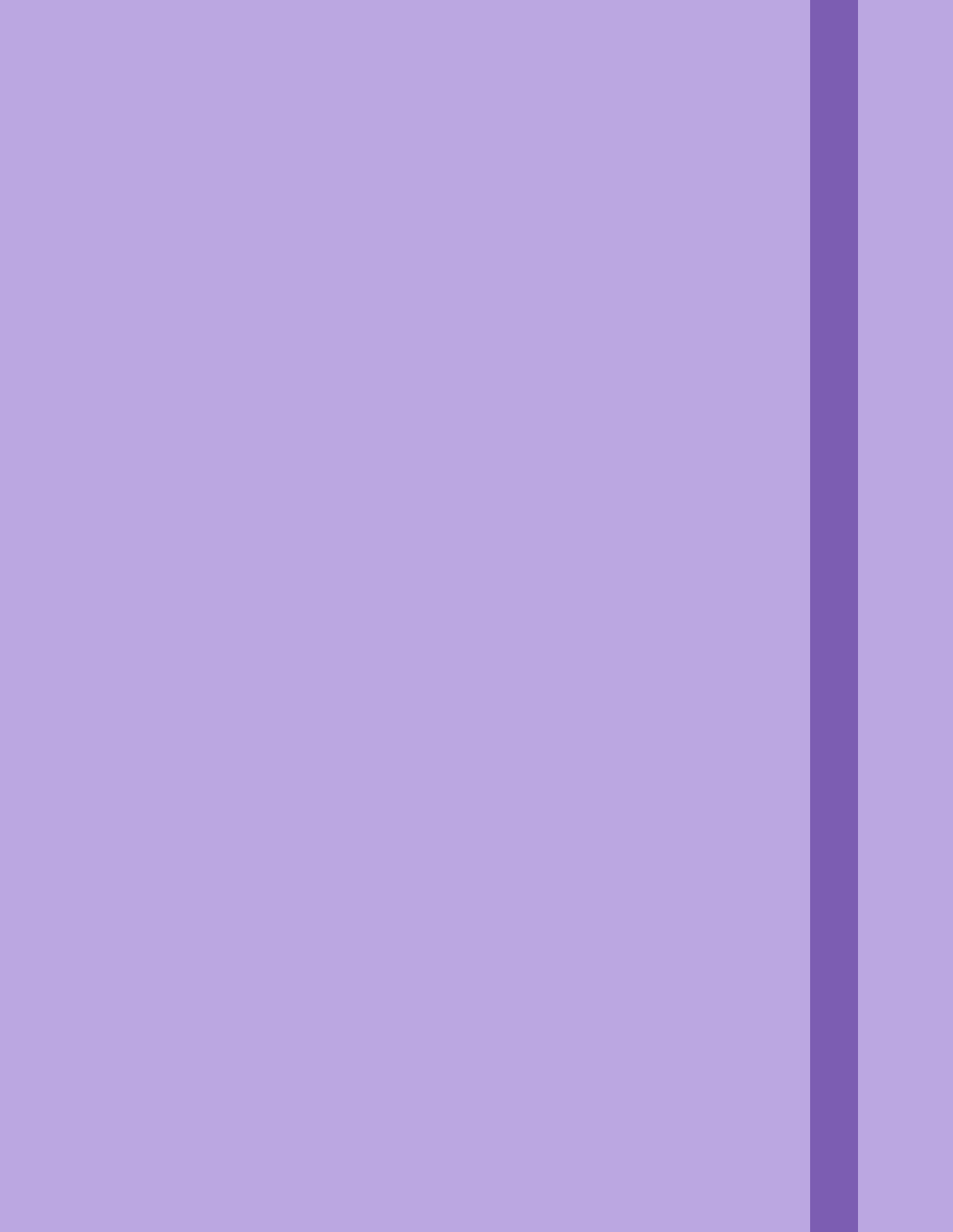




EX 1

WHAT IS THE EXPECTED NUMBER OF FLIPS
BEFORE WE SEE HTHT ?

EX 2

WHAT IS THE EXPECTED NUMBER OF FLIPS
BEFORE WE SEE HHTHH?

EX 3

GIVEN AN UNFAIR COIN SUCH THAT

$$P(H) = \frac{2}{3}$$

$$P(T) = \frac{1}{3}$$

WHAT IS THE EXPECTED NUMBER OF FLIPS
BEFORE WE SEE

- HTH

- HHT

EX 4

IMAGINE YOU HAVE A FAIR COIN THAT YOU Toss REPEATEDLY. TWO PLAYERS OBSERVE THE SEQUENCE OF TOSSES.

P1 WAITS FOR HHH
P2 WAITS FOR TTHH

THE ONE WHOSE PATTERN APPEARS FIRST WINS.

WHAT IS THE PROBABILITY THAT P1 WINS?

MORE EXERCISES

IN THE LECTURE NOTES YOU ALSO HAVE SOLUTIONS FOR THE NUMBER OF FLIPS TO

- HTTT#
- #TTT#TH
- HHTTH
- HTT#T
- HT#HT
- HTT#T
- #THT#T

BUT OF COURSE YOU CAN COME UP WITH A SEQUENCE YOURSELF!